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Learning Objectives

- Test sets of vectors for linear independence
- Determine if a set of vectors spans a given space
- Find a basis and dimension for null space and column space
- Apply the Rank-Nullity Theorem and Gram-Schmidt process

For subspace proofs, check all three axioms: zero vector, closure under addition, closure under scalar multiplication.

1. Are u and v linearly independent? Set up $cu + dv = 0$ and solve.

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

Answer: _____

2. Determine if the three vectors are linearly independent.

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \quad \vec{v}_3 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

Answer: _____

3. Do u and v span $\mathbb{R}^2$? Explain using the determinant.

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

Answer: _____

4. Is $W = \{(x, y, z) : x + y + z = 0\}$ a subspace of $\mathbb{R}^3$?

$$W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$$

Answer: _____



5. Find a basis and the dimension of the null space of A.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Answer: _____

6. Find rank(A) and nullity(A). Verify the Rank-Nullity Theorem.

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Answer: _____

7. Find a basis for the column space (range) of A.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}$$

Answer: _____

8. Use the Gram-Schmidt process to find an orthogonal basis for the span of $\vec{v}_1$ and $\vec{v}_2$.

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Answer: _____

9. Find the coordinate vector $[\vec{x}]_B$ of $\vec{x}$ with respect to basis B.

$$\vec{x} = (5, 3)^T, \quad B = \{(2, 1)^T, (1, 1)^T\}$$

Answer: _____

10. A linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is defined by $T(\vec{x}) = A\vec{x}$. Find $T([2, -1]^T)$.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Answer: _____





Problems 4–5: subspace check and null space basis. Problem 6 (Rank-Nullity): critical theorem, always verify $n = \text{rank} + \text{nullity}$. Problem 8 (Gram-Schmidt): show the projection formula explicitly.

Solutions

1. Are u and v linearly independent? Set up $cu + dv = 0$ and solve.

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

$$\rightarrow c[1, 2] + d[3, 6] = 0 \rightarrow c + 3d = 0, 2c + 6d = 0.$$

$$\rightarrow \text{Both equations: } c = -3d. \text{ Non-trivial solution exists (} d=1, c=-3 \text{).}$$

$$\rightarrow v = 3u \text{ — the vectors are parallel. Linearly dependent.}$$

Answer: Linearly dependent: $\vec{v} = 3\vec{u}$

2. Determine if the three vectors are linearly independent.

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \quad \vec{v}_3 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$\rightarrow \text{Form matrix } A = [\vec{v}_1 | \vec{v}_2 | \vec{v}_3] \text{ and compute } \det(A).$$

$$\rightarrow \det(\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 1 & 3 & 0 \end{bmatrix}) = 1(0+3) - 2(0+1) + 1(0-1) = 3 - 2 - 1 = 0.$$

$$\rightarrow \det=0 \rightarrow \text{linearly dependent.}$$

$$\rightarrow \text{Correction: } 1(1 \cdot 0 - (-1) \cdot 3) - 2(0 \cdot 0 - (-1) \cdot 1) + 1(0 \cdot 3 - 1 \cdot 1) = 1(3) - 2(1) + 1(-1) = 0.$$

$$\rightarrow \text{Result: } \det = 0 \rightarrow \text{vectors are linearly DEPENDENT.}$$

Answer: Linearly independent

3. Do u and v span $\mathbb{R}^2$? Explain using the determinant.

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$\rightarrow \det(\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}) = 3 + 2 = 5 \neq 0.$$

$$\rightarrow \text{Non-zero det means the columns are linearly independent.}$$

$$\rightarrow \{u, v\} \text{ is a basis for } \mathbb{R}^2. \checkmark$$

Answer: $\det = 5 \neq 0 \Rightarrow \{\vec{u}, \vec{v}\}$ spans $\mathbb{R}^2$

4. Is $W = \{(x, y, z) : x + y + z = 0\}$ a subspace of $\mathbb{R}^3$?

$$W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$$

$$\rightarrow 1) \text{ Zero vector: } 0+0+0=0 \checkmark.$$

$$\rightarrow 2) \text{ Closure under addition: if } x_1 + y_1 + z_1 = 0 \text{ and } x_2 + y_2 + z_2 = 0, \text{ then } (x_1 + x_2) + (y_1 + y_2) + (z_1 + z_2) = 0 \checkmark.$$

$$\rightarrow 3) \text{ Closure under scalar mult: } c(x) + c(y) + c(z) = c(x+y+z) = 0 \checkmark.$$

$$\rightarrow W \text{ satisfies all three subspace axioms.}$$

Answer: Yes, W is a subspace of $\mathbb{R}^3$



5. Find a basis and the dimension of the null space of A.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

→ Row reduce: already in echelon form. Pivots in cols 1 and 3.

→ Free variable: x_2 .

→ From row 2: $x_3 = 0$. From row 1: $x_1 + 2x_2 = 0 \rightarrow x_1 = -2x_2$.

→ Null space = $\{(-2t, t, 0) : t \in \mathbb{R}\}$. Basis: $\{[-2, 1, 0]^T\}$. $\dim = 1$.

Answer: $\text{Nul}(A) = \text{span}\{(-2, 1, 0)^T\}$, $\dim = 1$

6. Find $\text{rank}(A)$ and $\text{nullity}(A)$. Verify the Rank-Nullity Theorem.

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

→ A is 3×4 ($n=4$ columns). Already in echelon form.

→ Pivot columns: 1 and 3. Rank = 2.

→ Nullity = $n - \text{rank} = 4 - 2 = 2$.

→ Rank-Nullity: $2+2=4 \checkmark$.

Answer: rank = 2, nullity = 2, rank + nullity = 4 = n

7. Find a basis for the column space (range) of A.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}$$

→ Row reduce A to find pivot columns.

→ $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 3R_1$: $[[1, 2, 1], [0, 0, 1], [0, 0, 1]]$.

→ $R_3 \leftarrow R_3 - R_2$: $[[1, 2, 1], [0, 0, 1], [0, 0, 0]]$. Pivots in cols 1 and 3.

→ Basis for $\text{Col}(A)$ = original pivot columns = $[1, 2, 3]^T$ and $[1, 3, 4]^T$.

Answer: $\text{Col}(A) = \text{span}\{(1, 2, 3)^T, (1, 3, 4)^T\}$

8. Use the Gram-Schmidt process to find an orthogonal basis for the span of v_1 and v_2 .

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

→ $u_1 = v_1 = [1, 1, 0]^T$.

→ $\text{proj}_{\{u_1\}}(v_2) = (v_2 \cdot u_1 / u_1 \cdot u_1) u_1 = (1/2)[1, 1, 0]^T = [1/2, 1/2, 0]^T$.

→ $u_2 = v_2 - \text{proj} = [1, 0, 1]^T - [1/2, 1/2, 0]^T = [1/2, -1/2, 1]^T$.

→ Verify: $u_1 \cdot u_2 = 1/2 - 1/2 + 0 = 0 \checkmark$.

Answer: $\vec{u}_1 = (1, 1, 0)^T$, $\vec{u}_2 = (1/2, -1/2, 1)^T$



9. Find the coordinate vector $[x]_B$ of x with respect to basis B .

$$\vec{x} = (5, 3)^T, \quad B = \{(2, 1)^T, (1, 1)^T\}$$

$$\rightarrow \text{Solve } c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}.$$

$$\rightarrow 2c_1 + c_2 = 5 \text{ and } c_1 + c_2 = 3.$$

$$\rightarrow \text{Subtract: } c_1 = 2. \text{ Then } c_2 = 1.$$

$$\rightarrow [x]_B = [2, 1]^T.$$

Answer: $[\vec{x}]_B = (2, 1)^T$

10. A linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is defined by $T(x) = Ax$. Find $T(\begin{bmatrix} 2 \\ -1 \end{bmatrix})$.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\rightarrow T(x) = Ax = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

$$\rightarrow \text{Row 1: } 1(2) + 2(-1) = 0.$$

$$\rightarrow \text{Row 2: } 3(2) + (-1)(-1) = 7.$$

$$\rightarrow T(\begin{bmatrix} 2 \\ -1 \end{bmatrix}) = \begin{bmatrix} 0 \\ 7 \end{bmatrix}.$$

Answer: $\begin{bmatrix} 0 \\ 7 \end{bmatrix}$

